

# Artificial Intelligence

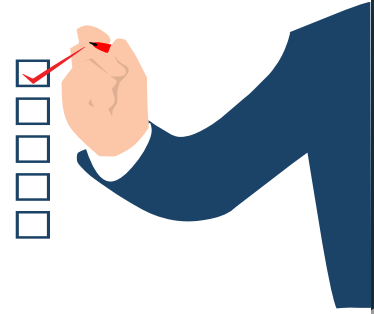


# ChatGPT

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## Work Plan

1. Introduction to AI
2. Mathematics for AI
3. Data Preprocessing
4. Supervised Learning Algorithms
5. Unsupervised Learning Algorithms
6. Reinforcement Learning
7. Neural Networks & Deep Learning
8. Natural Language Processing (NLP)
9. Model Evaluation and Metrics
10. Advanced Topics

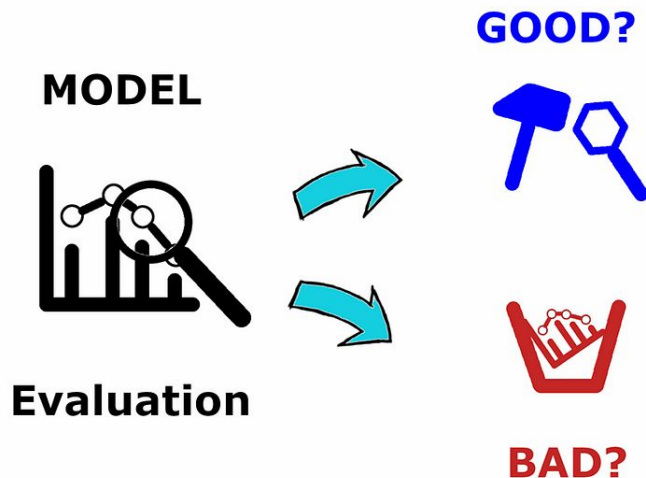


# Model Evaluation and Metrics

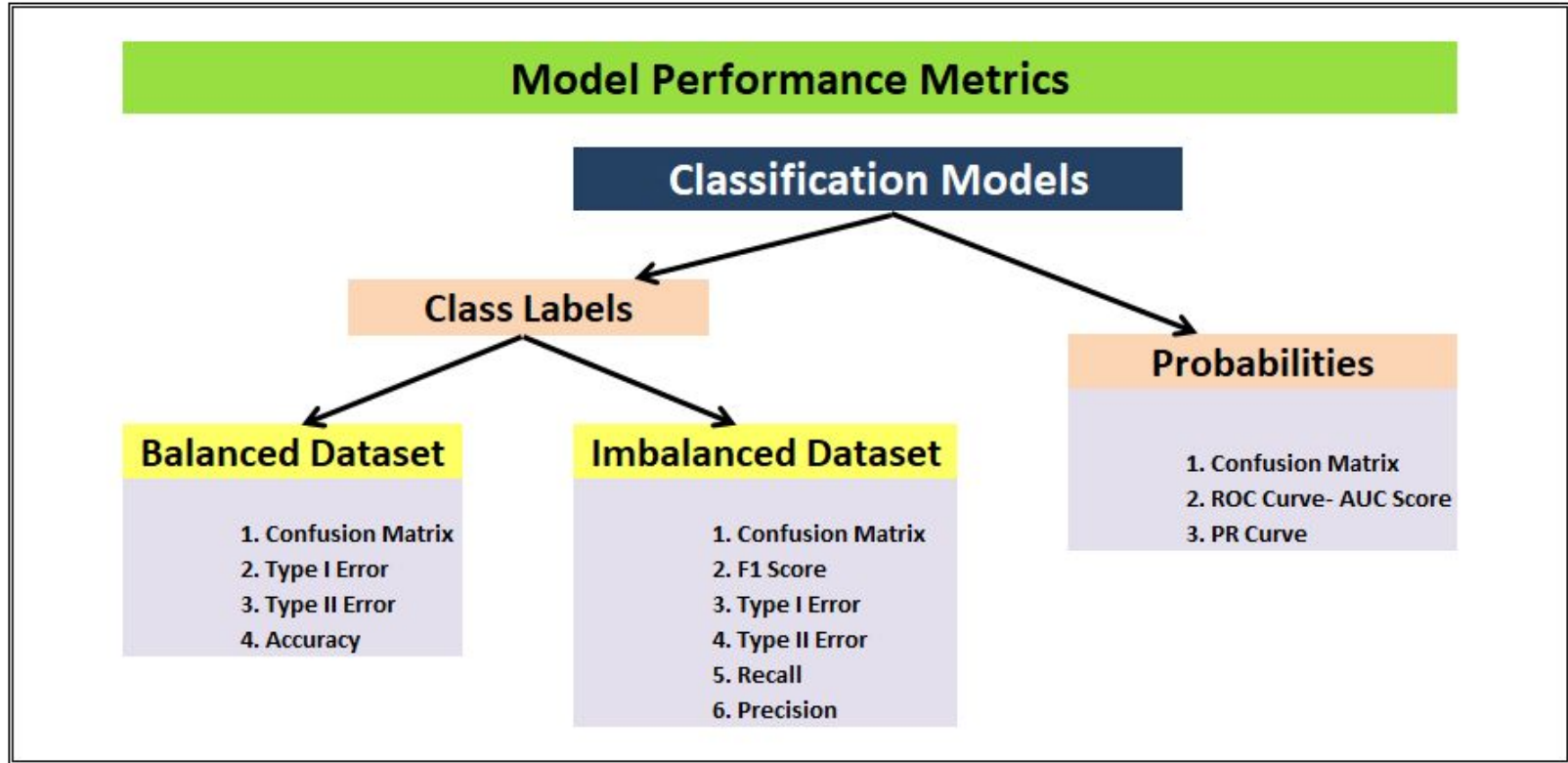


## Model evaluation

- Helps us check how well our machine learning models are **performing**.
- We use different metrics depending on whether it's a classification or regression problem.



## Model evaluation



## Performance Metrics

|                  |                 | <i>Actual</i>                                                       |                                                                               |
|------------------|-----------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|
|                  |                 | <i>Positive</i>                                                     | <i>Negative</i>                                                               |
| <i>Predicted</i> | <i>Positive</i> | <b>True Positive</b><br><i>Predicted has cancer<br/>Has Cancer</i>  | <b>False Positive</b><br><i>Predicted has cancer/Does<br/>not have cancer</i> |
|                  | <i>Negative</i> | <b>False Negative</b><br><i>Predicted not cancer<br/>Has cancer</i> | <b>True Negative</b><br><i>Predicted not cancer<br/>Does not have cancer</i>  |

## Performance Metrics : TP, TN, FP, FN

TP ( True Positive ) : Model **correctly predicted Positive**, e.g., model says *spam*, and it is **spam**

TN ( True Negative ) : Model **correctly predicted Negative**, e.g., model says *not spam*, and it is **not spam**

## Performance Metrics : TP, TN, FP, FN

FP ( False Positive ) (Type I Error) : Model predicted **Positive**, but it's **actually Negative**,

ee.g., model says *spam*, but it's **not spam**

FN ( False Negative ) (Type II Error) : Model predicted **Negative**, but it's **actually Positive**,

e.g., model says *not spam*, but it **is spam**

# **Classification Metrics**

**Classification Metrics** : Used when your model predicts categories or classes (e.g., spam or not spam).

- Accuracy
- Precision
- Recall (Sensitivity)
- F1-Score
- Confusion Matrix
- ROC Curve (Receiver Operating Characteristic)
- AUC (Area Under the Curve)

**Accuracy** : How often the model is correct

**Formula** :

- Correct Predictions / Total Predictions
- $( TP + TN ) / ( TP + FP + TN + FN )$

## Precision

- Out of all predicted positives, how many were actually positive ?
- Used when false positives are costly
- Formula :  $TP / (TP + FP)$

## Recall (Sensitivity)

- Out of all actual positives, how many were predicted correctly ?
- Used when missing positives is costly
- Formula:  $TP / TP + FN$

## F1-Score

- Balance between precision and recall
- Formula :  $2 * ( \text{Precision} * \text{Recall} ) / ( \text{Precision} + \text{Recall} )$

## Confusion Matrix

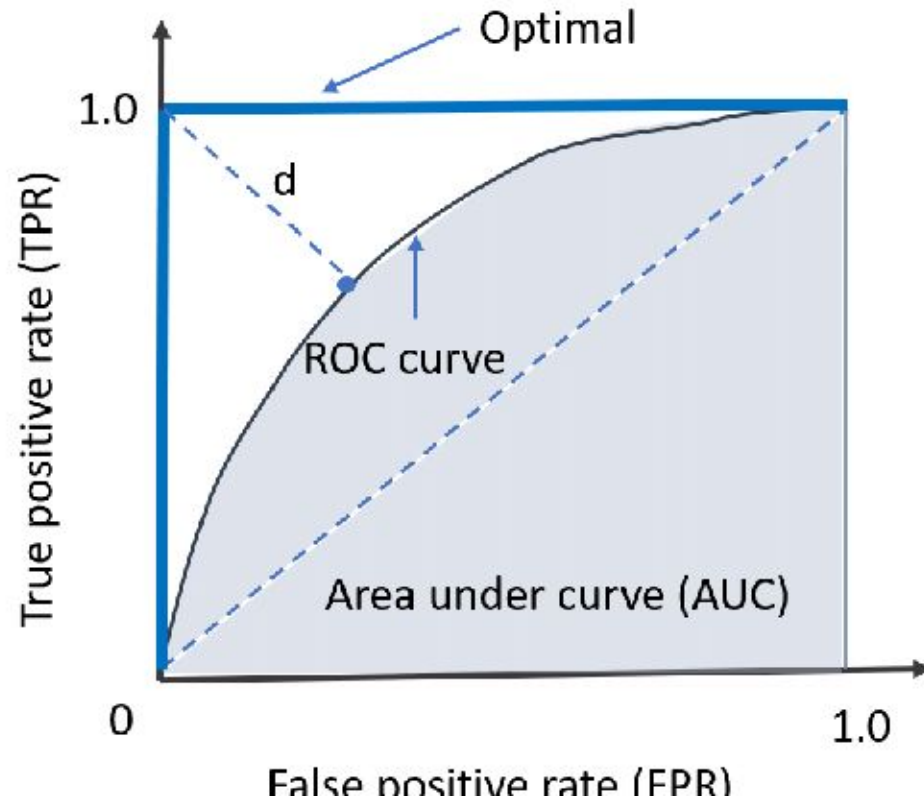
A table showing True Positives, False Positives, False Negatives, and True Negatives.

| n=165          | Predicted:<br>NO | Predicted:<br>YES |
|----------------|------------------|-------------------|
| Actual:<br>NO  | 50               | 10                |
| Actual:<br>YES | 5                | 100               |

## ROC Curve (Receiver Operating Characteristic)

- Shows the trade-off between True Positive Rate and False Positive Rate.
- To evaluate the performance of a **binary classifier** across all classification thresholds.
- **Axes :**
  - **X-axis** : False Positive Rate (FPR) =  $FP / (FP + TN)$
  - **Y-axis** : True Positive Rate (TPR) =  $TP / (TP + FN)$  = **Recall**

## ROC Curve (Receiver Operating Characteristic)



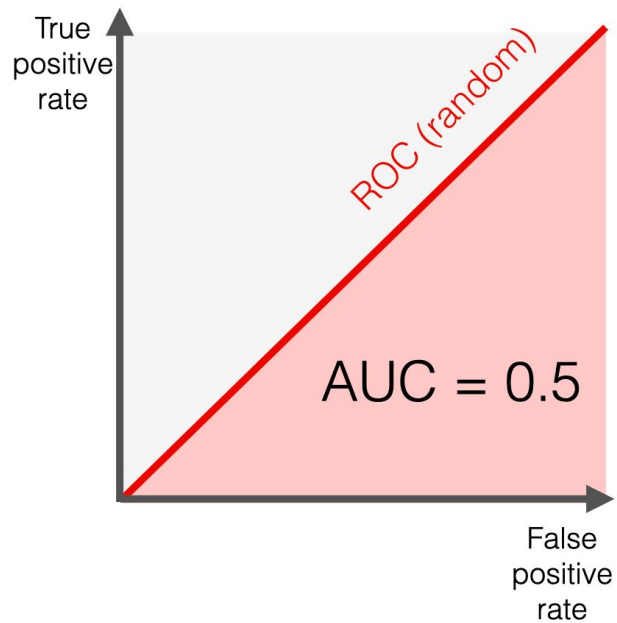
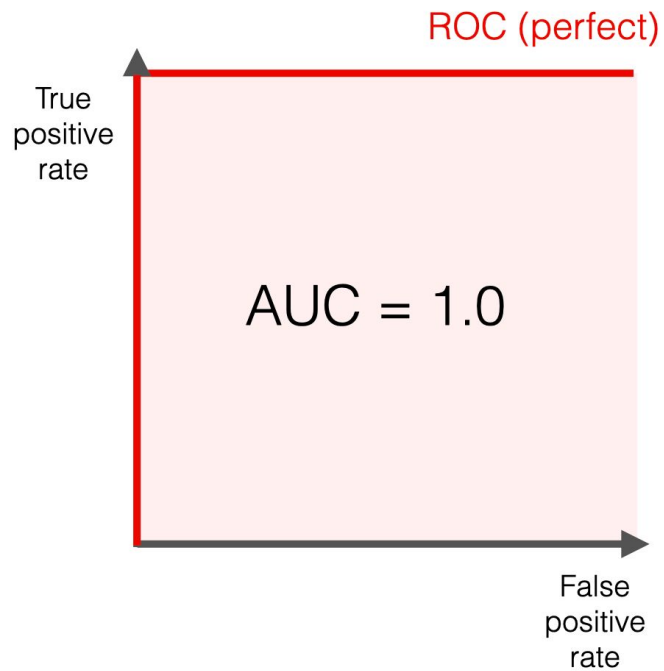
## AUC (Area Under the Curve)

- AUC = **Probability** that the classifier ranks a **random positive** example **higher** than a random negative one
- It quantifies **how well the model separates classes**
- Measures the entire area under the ROC curve

## AUC (Area Under the Curve)

- **Higher** AUC = **better** performance
- If a model has **AUC = 0.85**, it means : There's an 85% chance the model ranks a randomly chosen **positive** sample higher than a randomly chosen **negative** one.

## AUC (Area Under the Curve)



# Regression Metrics

## Regression Metrics

Used when the model predicts continuous values ( like house price, temperature )

- Mean Absolute Error ( MAE )
- Mean Squared Error ( MSE )
- R-Squared (  $R^2$  )
- Adjusted R-Squared

**Mean Absolute Error ( MAE )** : Average of the absolute errors

summation of all values  
(with i ranging from 1  
to n)

this operator gives the  
absolute value of a  
number

$$\text{MAE} = \frac{\sum_{i=1}^n |y - \hat{y}_i|}{n}$$

No. of data  
points

**y = actual value,  $\hat{y}$  = predicted value**

**Mean Squared Error (MSE)** : Average of squared errors. Penalizes large errors more than

MAE

number of  
samples

real  
value

predicted  
value

$$\sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

sum of the errors of all samples

The diagram illustrates the Mean Squared Error (MSE) formula. It features the mathematical expression  $\sum_{i=1}^n (Y_i - \hat{Y}_i)^2$ . Above the summation symbol is a blue box containing the variable  $n$ , with the text 'number of samples' in blue above it. The term  $Y_i$  is enclosed in a green box, with 'real value' in green above it. The term  $\hat{Y}_i$  is enclosed in a red box, with 'predicted value' in red above it. A horizontal curly brace is positioned below the entire expression, with the text 'sum of the errors of all samples' in grey below the brace.

**R-Squared (  $R^2$  )** : Explains how much variance in the output is explained by the model.

Ranges from 0 to 1 ( higher is better )

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

**Adjusted R-Squared** Like  $R^2$ , but adjusts for the number of predictors (features). Useful when comparing models with different numbers of features.

$$\text{Adjusted } R^2 = 1 - \frac{(1 - R^2)(N - 1)}{N - p - 1}$$

Where

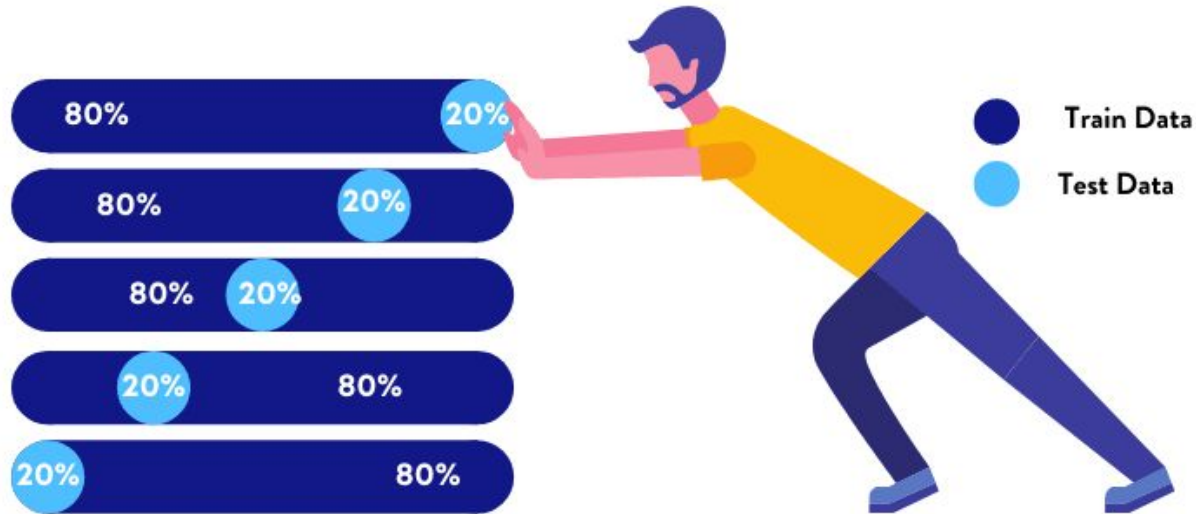
$R^2$  Sample R-Squared

$N$  Total Sample Size

$p$  Number of independent  
variable

**P : The number of independent variables (features) / Predictors in your regression model**

# Cross Validation



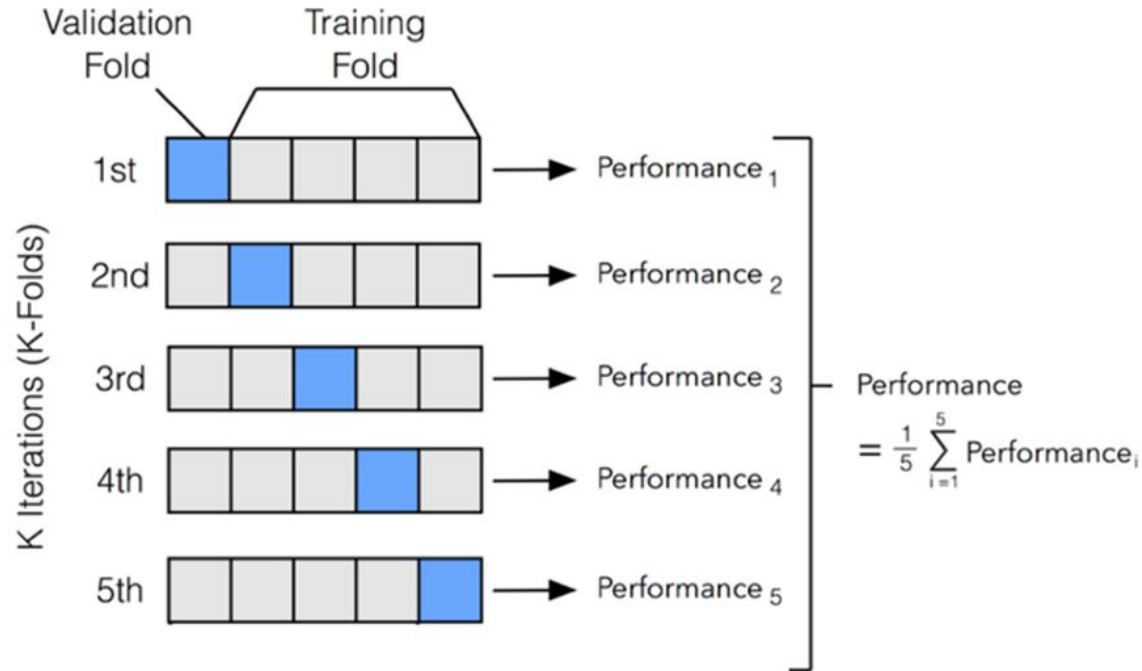
[dataaspirant.com](http://dataaspirant.com)

## Cross validation

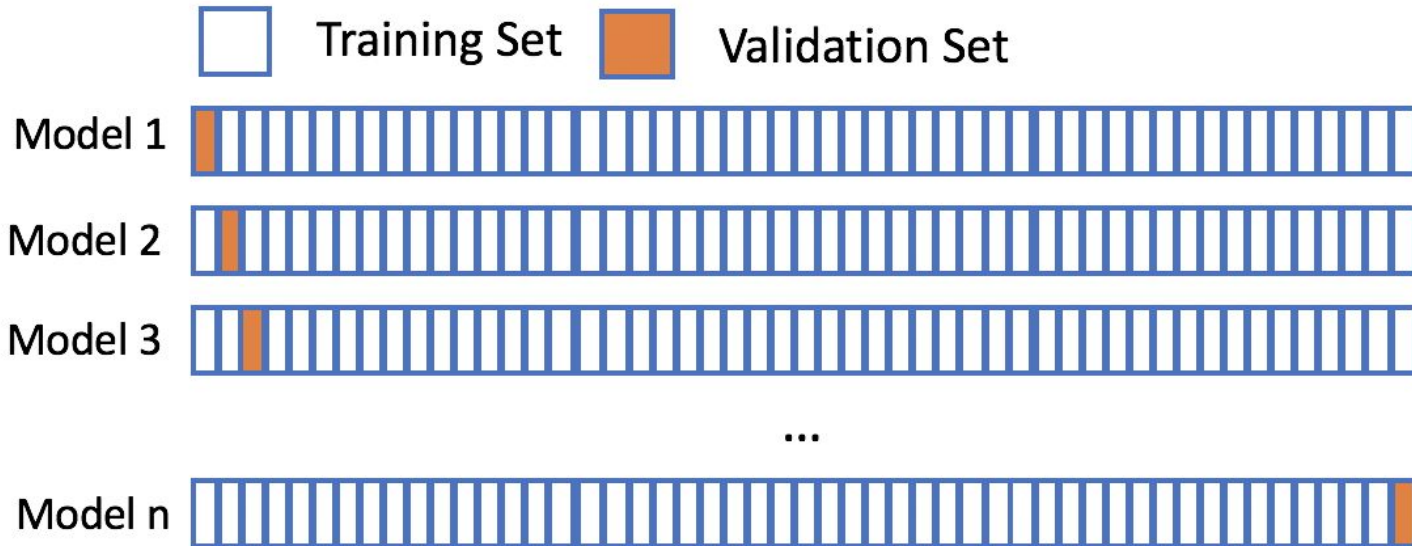
Used to test model performance on different splits of the data

- K-Fold Cross-Validation
- Leave-One-Out Cross-Validation (LOOCV)
- Stratified K-Fold

**K-Fold Cross-Validation** Split data into k equal parts. Train on k-1 and test on the remaining part. Repeat k times.

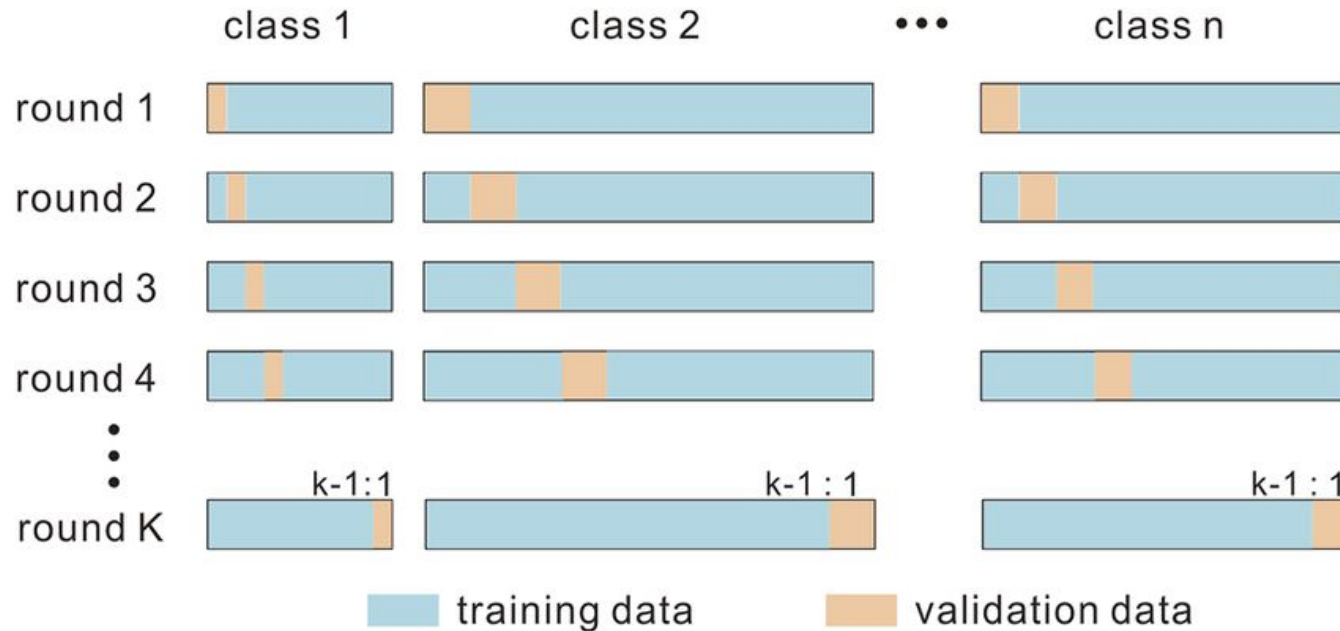


**Leave-One-Out Cross-Validation (LOOCV)** A special case of K-Fold where  $k = \text{number of data points}$ . Very slow but thorough.



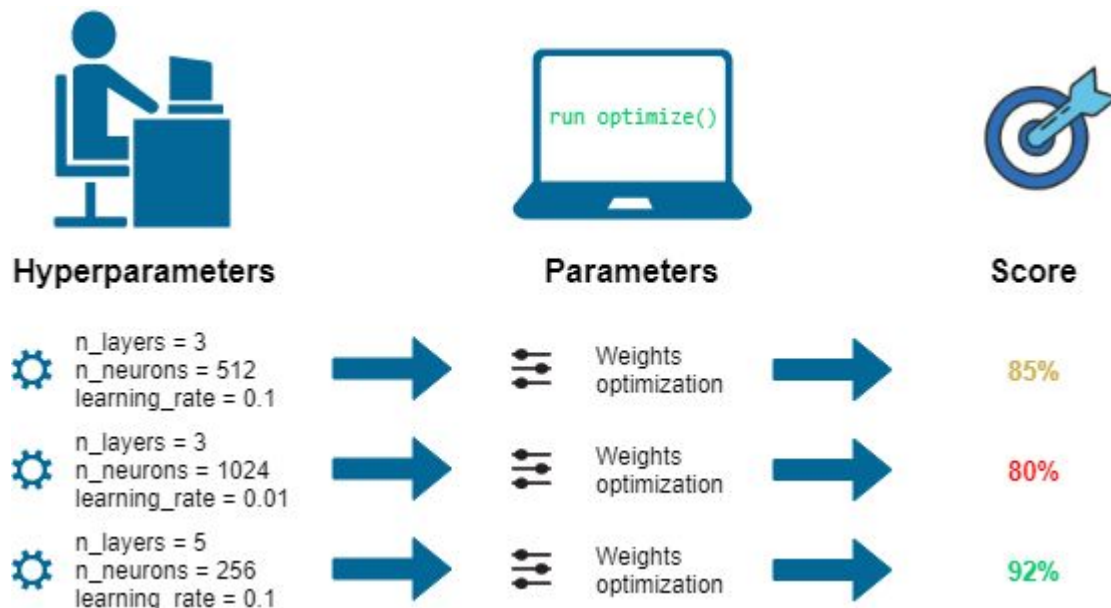
**Stratified K-Fold** Same as K-Fold, but keeps the ratio of classes the same in each fold.

Useful for imbalanced datasets

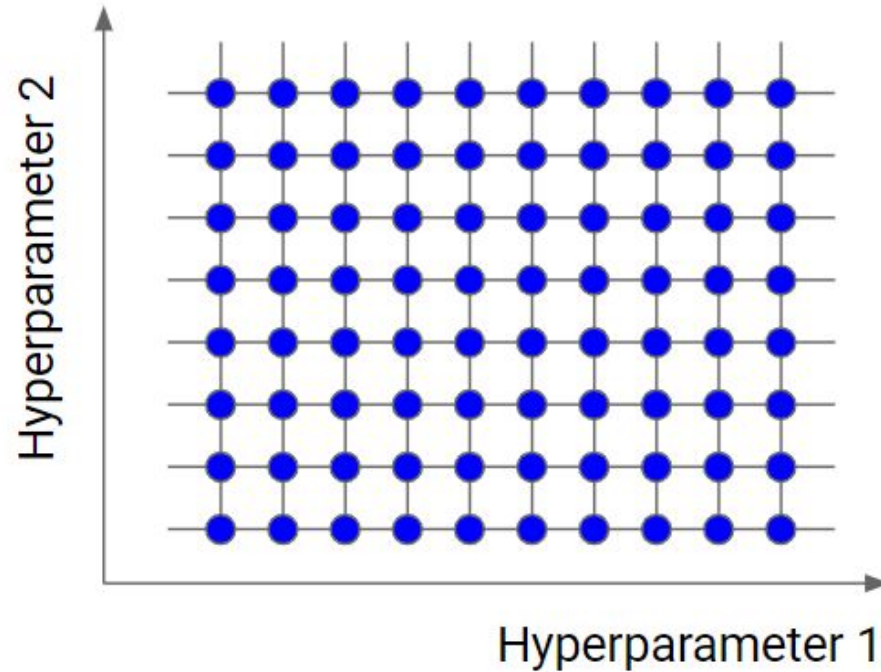


# Hyperparameter Tuning

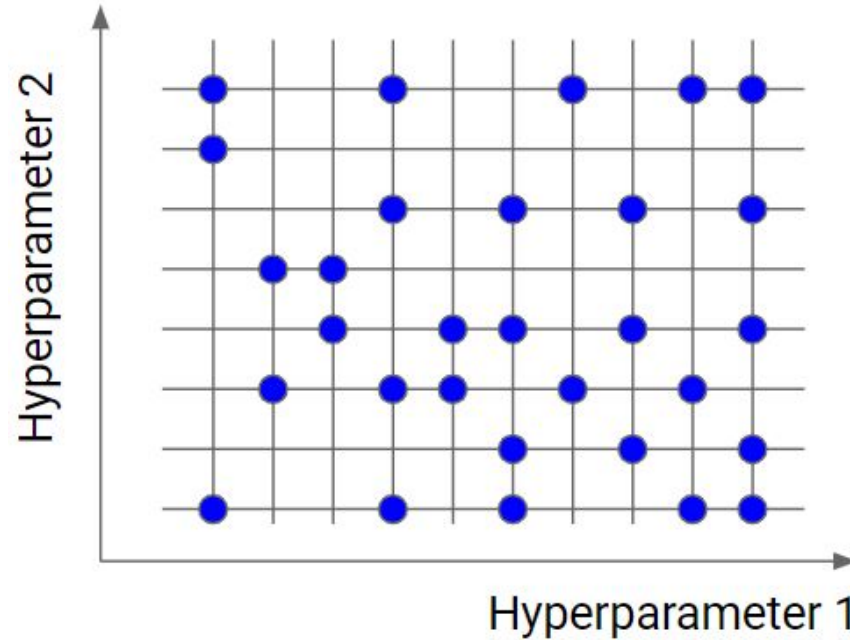
**Hyperparameters** are settings that control how a model learns (like learning rate, depth of a tree, etc.).



**Grid Search** Tests all combinations of given hyperparameter values.

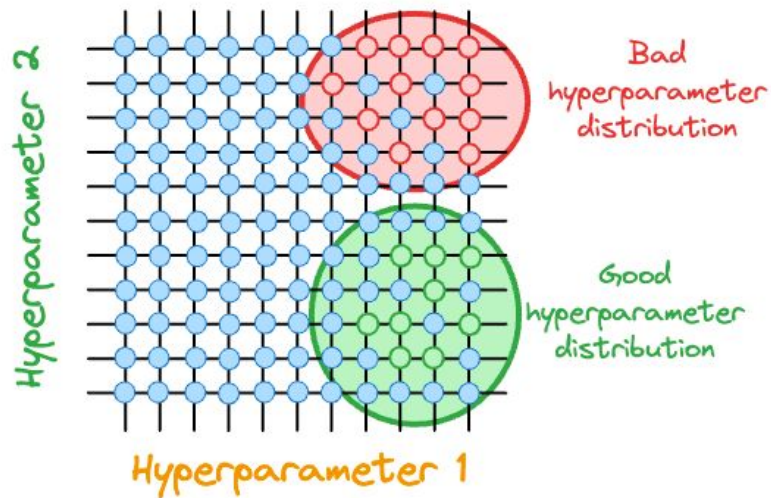


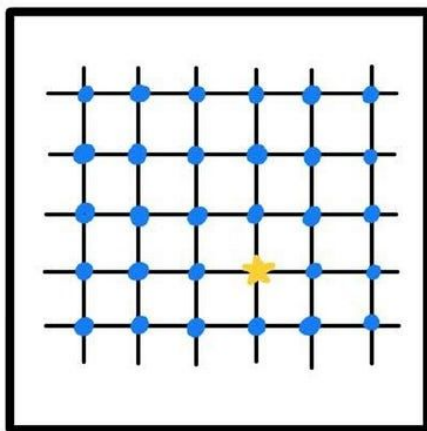
**Random Search** Randomly selects combinations. Faster than Grid Search.



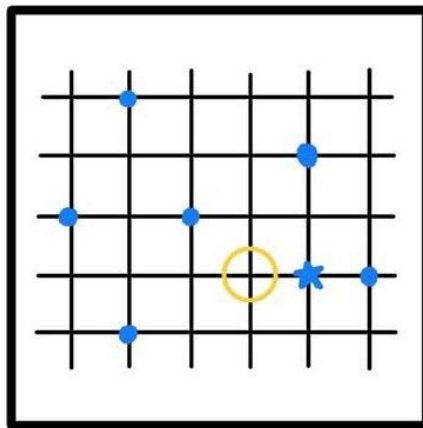
**Bayesian Optimization** Uses past results to pick the next best combination. Smart and efficient

## Bayesian Optimization

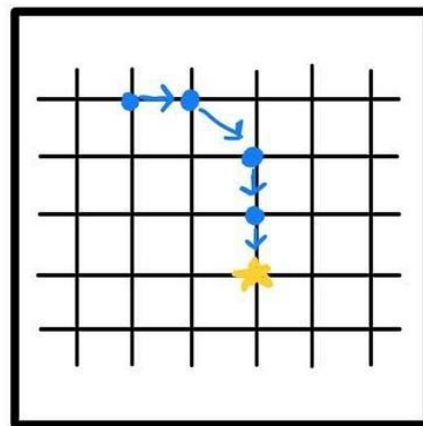




Grid Search



Random Search



Bayesian  
Optimization

- Evaluation points
- ★ Optimal parameters
- ★ Local optimal parameters

